HONG KONG BAPTIST UNIVERSITY

Audio and Speech Processing

Practical Assignment I

(100 marks | 10%)

<u>Introduction</u>

This practical assignment is designed to assess students' understanding and programming skills of conducting Machine Learning with Audio Signals. Students will be tested on their replication and implementation of building a basic pipeline, including *data preprocessing, analysis,* and *feature extraction (engineering)*. Bonus scores will be provided to students for their innovative programming on an extra task with additional feature construction. They are also required to report results by showing critical thinking in interpreting the results.

Data

Like the sample data demonstrated in the lab, two sets of data are provided. Each student will work on the data according to the number of the last digit in your student ID. Please refer to *Table 1 in Appendix A* to find the data to work on.

Program Code Template

The basic program codes by Python are provided for your reference as in the document "*audio_1.ipynb*". All the tasks can be implemented based on the given codes with slight adjustments of commenting certain functions, changing parameters or features. Google Colab and PyCharm can be the platform for you to execute the programing.

Tasks

- **Basic Task**: Replicate the given codes in the template file and successfully run the codes on your given subset of data. Report the implementation results in your *.docx* file. Provide basic descriptions of the results.
- **Task 1**: Sampling frequency effects. Change the sampling frequency *sr* to 44100 Hz when generating the spectrogram and waveform. Comment on the difference between the figures. Report how it changes audio sound play effect. Include the changed waveform and spectrogram figures in the report.
- **Task 2**: Spectrogram features. Compare the spectrogram features when the window lengths are 128, 256, and 512. Record these spectrogram images, and the related size. Fix the window length to be 256, change the *hop_length* as ½ and ¼, record the spectrogram images and the sizes. Plot the figures with the chosen musical instruments. Report how these changes affect visualization.
- **Task 3**: Data augmentation. Compare audio effect and spectrogram when the function *librosa.effects.trim* is applied to the signals. Report your findings.
- **Task 4**: MFCC features. Compare the Mel spectrogram when the number of MFCC is 12, 24, and 32. Record the sizes of features, and discuss the difference in the report.
- ***Extra/bonus Task 1**: Re-express each task code above as function format. Specify the inputs and outputs of the functions. Report your results in report and "*code_improved.ipynb*".
- ***Extra/bonus Task 2**: Perform audio dimensional reduction with down sampling. Perform *PCA* in *sklearn* for dimensional reduction on the given audio spectrogram data. Discuss the implementation results. Check *sklearn* document to call the function correctly.

Report Format

- Indicate the following information in your report:
    - Course Code & Title
    - Practical Assignment I
    - Summary Report on Audio Processing and Analysis
    - Student Name and ID
    - Date of Submission
- **Tasks and Results**: Report the results to each task (with point-wise subsections, e.g., 1.1, 1.2, 1.3…). Provide results descriptions and interpret them with proper analysis. Shows your understanding of audio features. It will make the report outstanding.
- **Summary**: Summarize interesting findings observed in the task.
- **Discussion**: Discuss the interesting and difficult aspects in implementation.
- **Conclusion**: Conclude the report and provide suggestions in implementation.
- **References**: List of references (APA) and online resources (if any)

Submission

1. Submit ONE *.ipynb* file containing all your program codes for tasks. Rename the. ipynb file to your name, e.g. ChanTaiMan.*ipynb*. Submit the bonus problem files separately.
2. Submit ONE word file reporting your results and findings. Rename the *.docx* file to your name, e.g. ChanTaiMan.*docx*.

Marking Criteria

Items	Evaluation Standard	Marks
Basic Task	Replication of running codes on given data	10
Task 1	Success in running task 1 and report the result	15
Task 2	Success in running task 2 and report the result	15
Task 3	Success in running task 3 and report the result	15
Task 4	Success in running task 4 and report the result	15
Report	Organized, elaborative, well-formatted	15
Analysis	In-depth thinking and reflections	15
Total:		100
Bonus:	Innovative programing and analytical skills	30

Appendix A

You should work on one of the corresponding subset data as listed in Table 1.

Last digit of student ID	Name of subset data
0-4	10Piano.wav, 10Violins.wav
5-9	10String.wav, 10Bass.wav

Table 1 - Mapping of Student ID and Subset Number.